MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, DETAIL LIGHTING FOR BOILER OUTER CASING

100 WATTS, 120 VOLTS, WATERTIGHT, SYMBOL 159.1

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

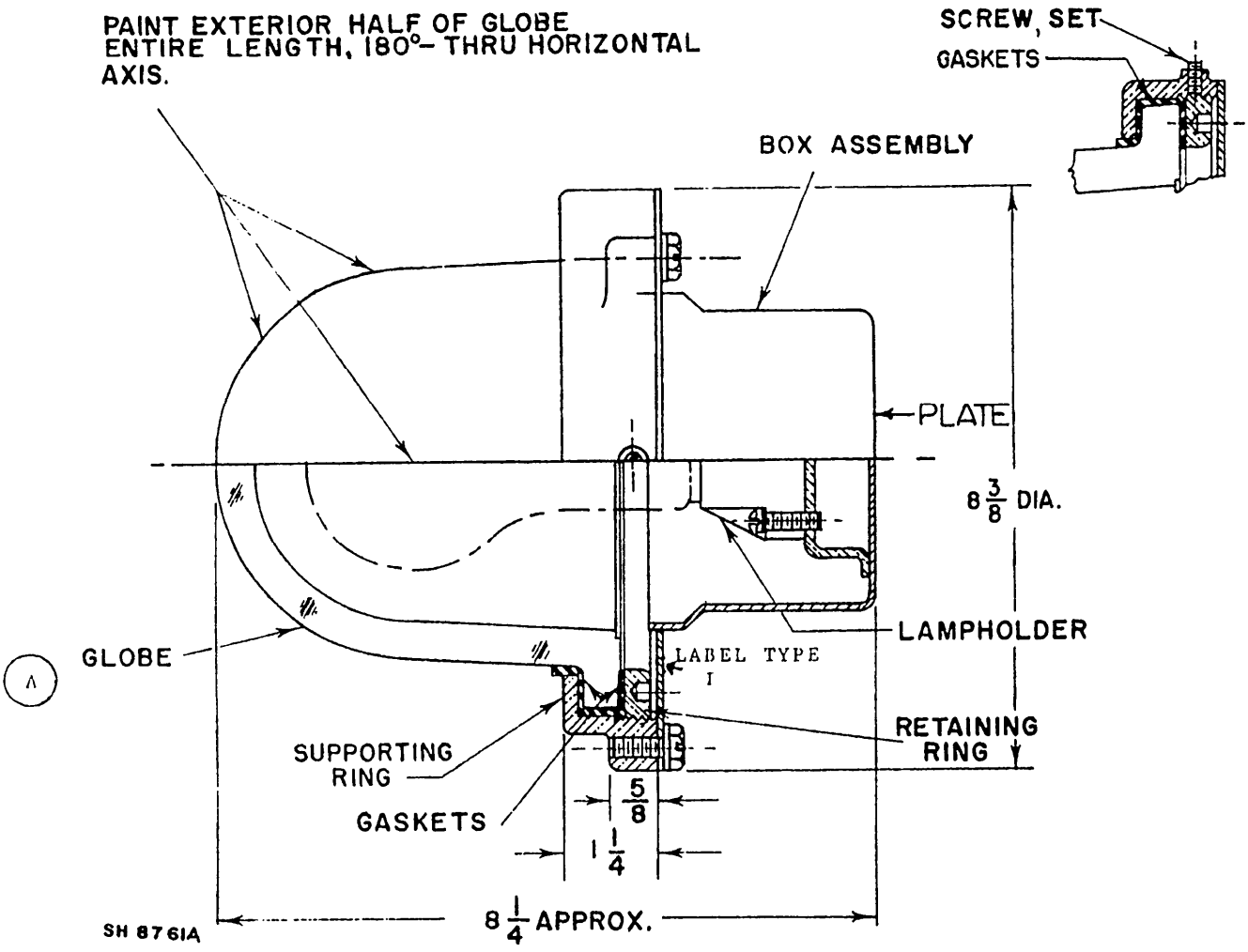


FIGURE 1. Typical view of Symbol 159.1

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(Λ) denotes changes.

FSC 6210

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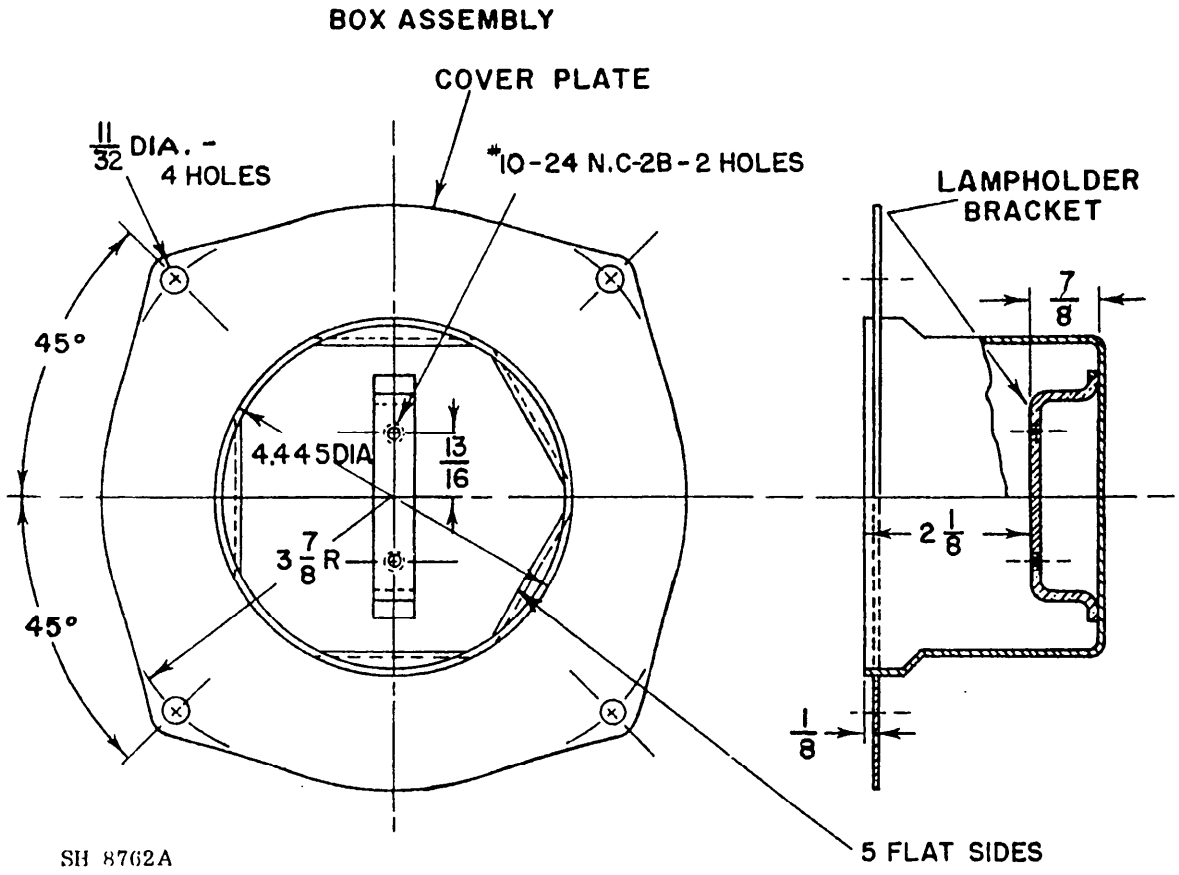


FIGURE 2. Box assembly.

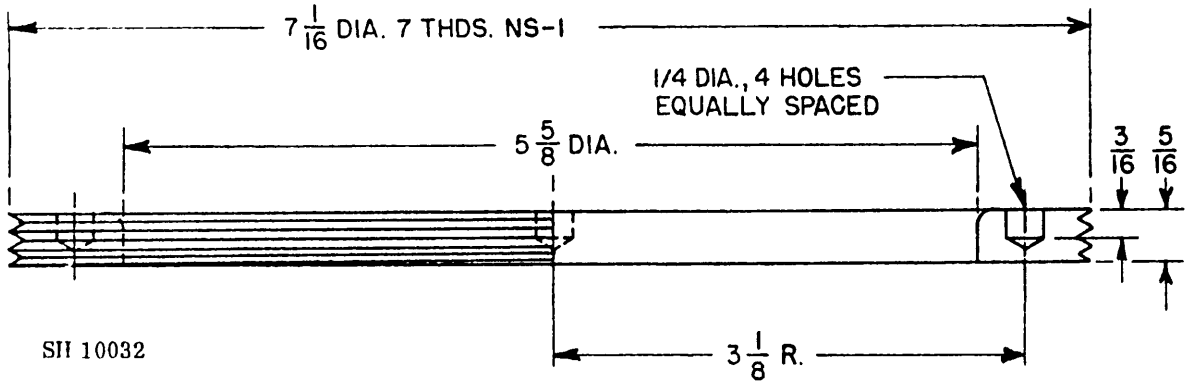
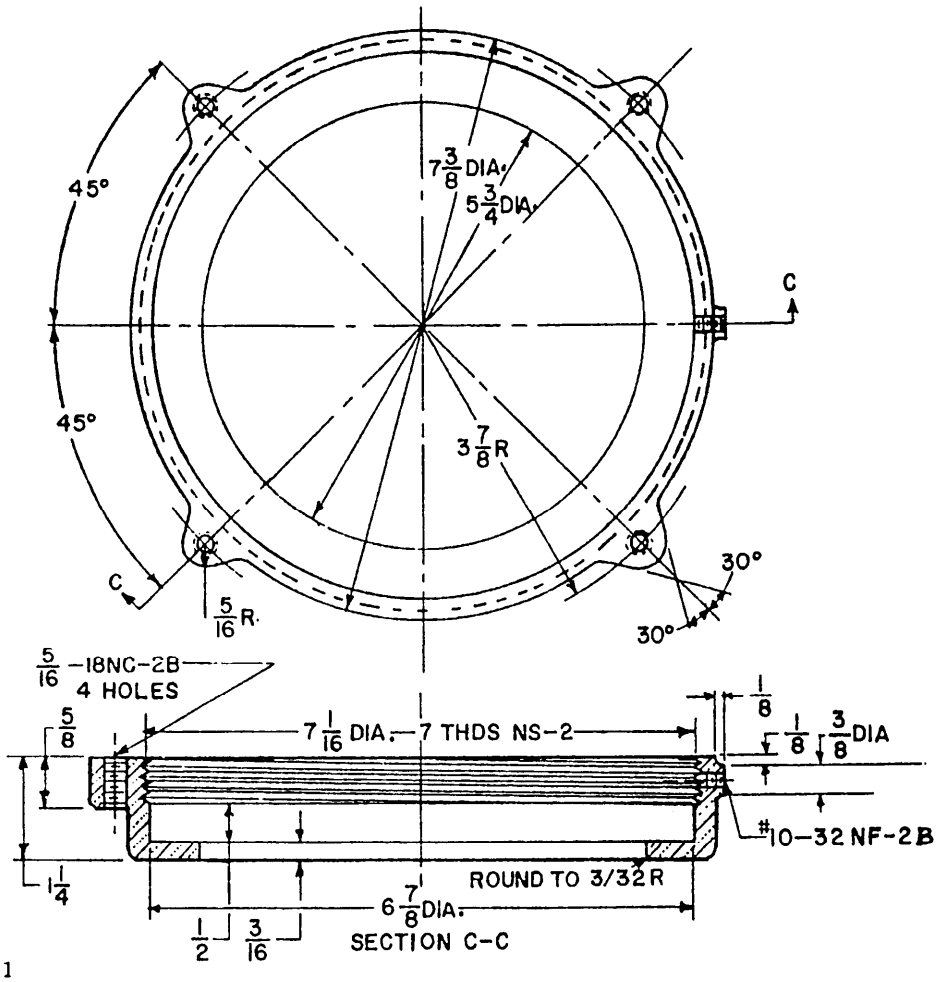


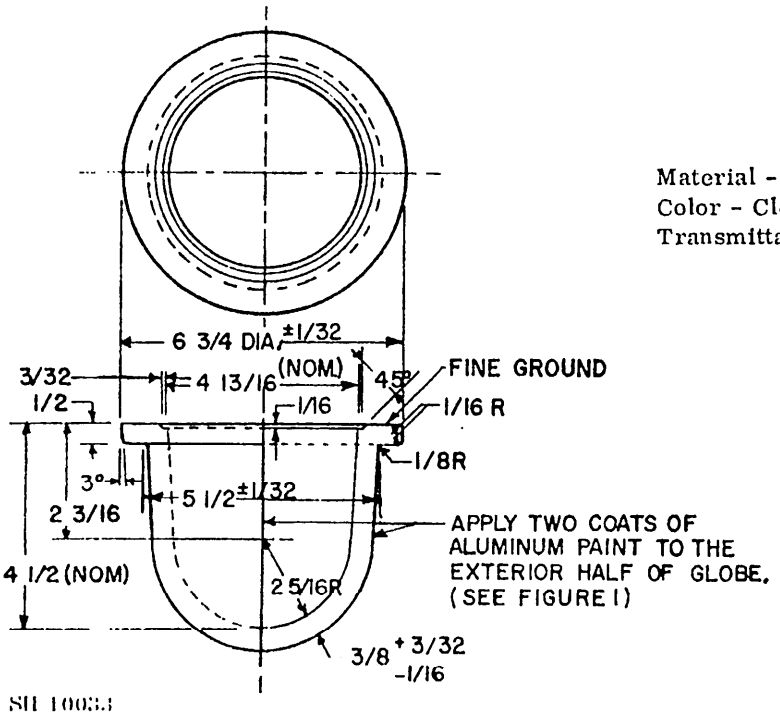
FIGURE 3. Retaining ring.

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SH 10031

FIGURE 4. Supporting ring.



Material - High impact, heat resistant, glass.
Color - Clear transparent
Transmittance - 85 percent minimum.

SH 10033

FIGURE 5. Globe.

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REQUIREMENTS:

CLASSIFICATION:
Type II
Class:I

SYMBOL NUMBER:
159.1: Fixture, detail lighting for boiler outer casing, for use with one 100-watt white lamp.

DESIGN:
Electrical rating: 120 volts, 100 watts.
Dimensions and configuration: See figures 1 through 5.
Box assembly, including cover plate, and lampholder bracket: Brass, see figure 2.
Supporting ring: Brass, or the following alternate materials may be used:
Aluminum bronze.
Tin nickel bronze.
Phosphor bronze.
Metal used shall be of a type which may be arc welded.
Gaskets: Asbestos. The portion of the glass globe in contact with metal parts shall be completely gasketed as shown on figure 1.
Set screw: Brass, or corrosion resistant steel, headless, #10-32, 3/8 inch long (see figure 1).
Retaining ring: Brass, bronze, or corrosion resistant steel.
Lampholder: MIL-L-970/6, two contact.
Globe: Figure 5. Light shielding shall be accomplished by the application of two coats of aluminum paint to the exterior half of the globe as shown on figures 1 and 5.
Weight: 8 pounds, max.
Lamp: This fixture shall be designed to accommodate a 100 watt lamp conforming to W-L-101/77, Industry Number 100A/RS(120) (lamp is not furnished). (A)
Identification plate: One plate shall be furnished in accordance with MIL-F-16377 applied as shown on figure 1. The information to be shown on the plate shall be the information required for a type I label in accordance with MIL-F-16377. (A)

QUALITY ASSURANCE:	Quality assurance shall be as specified in MIL-F-16377 and as follows
First article inspection:	
Complete fixtures:	See table I
Globes:	See table II
Quality conformance:	
Comparison inspection:	Same as the first article
Inspection of product for delivery:	
Examination:	See MIL-F-16377
Examination for preparation of delivery:	See MIL-F-16377
Quality conformance tests:	See MIL-F-16377

GENERAL INFORMATION: Assignment of Military part numbers and general information is shown in table III.

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TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
∧ Enclosure effectiveness	3.5.11	4.8.14	Totally encl.
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	5.8.8	
∧ Enclosure effectiveness	3.5.11	4.8.14	Totally encl.
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Mechanical shock	3.7.5.6	4.8.17.3	8 ft. - lbs.
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Optical uniformity	3.7.5.1	4.8.4	
Paint removal	3.7.5.5	4.8.17.2	
Color and luminous transmittance	3.7.3.7	4.8.6.3	
Magnetic permeability	3.4.8	4.8.18	

1/ The supporting ring shall be welded to a test plate which in turn shall be securely attached to shock machine for bulkhead mounting.

TABLE II. First article and comparison inspection for globes.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
∧ Enclosure effectiveness	3.5.11	4.8.14	Totally encl.
Mechanical shock	3.7.5.6	4.8.17.3	8 ft.-lbs.
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Paint removal	3.7.5.5	4.8.17.2	
Color and luminous transmittance	3.7.3.7	4.8.6.3	

TABLE III. Part number and general information.

Military Part number M16377/29-	Item name	National Stock Number (NSN)	Remarks
159.1	Complete fixture, Symbol 159.1	6210-00-069-2033	1/
001	Globe		2/ 3/

1/ Formerly covered on Drawing 9000-S6401-73989 and MS17785.

2/ Formerly piece number 15 of MS17785.

3/ Same as M16377/25-002 of MIL-F-16377/25, except painted as specified herein.

Preparing activity:
Navy - SH
(Project 6210-N577)

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